Decline Report 2022: Paths, Distribution, and Deflation

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Abstract

China's prolonged downturn that began in the third quarter of 2022 has been marked by low inflation and can be identified as a deflationary crisis, with wide-ranging social consequences that point to deeper systemic problems. This essay argues that the structural factors behind the current deflation are a growth path long dependent on the demographic dividend, a development model led by exports and investment, the systematic suppression of labor compensation and public services, and the intertwined expansion of land finance and local implicit debt, which together produced a weak consumption base and chronically insufficient domestic demand. Using cross-country data on 77 economies from 2011 to 2021, the essay empirically tests the effect of the distribution structure on deflation risk; the regression results support the hypothesis that distributional imbalance induces economic contraction. The essay also builds a simplified three-sector New Keynesian DSGE model with heterogeneous households to simulate the response of the price level to an improvement in the distribution structure, and finds that raising the economic position of middle- and low-income groups helps relieve deflationary pressure. Finally, the essay discusses policy paths that operate through distributional adjustment and points to directions for further thought. This essay is dedicated to the dull life of an age of recession.

Keywords: deflation; distribution structure; demographic dividend; DSGE; economic crisis

1 The Crisis

1.1 Signs

This is not merely "the end of the era of growth": our economy is not standing still but sinking into recession. A cultural-studies and sociological perspective allows me to observe that the cultural atmosphere around us has already shifted. Anti-involution sentiment, the rejection of meritocracy, individual liberation, and pluralism are the positive side of this shift; intensifying nationalism and conservatism, recurrent anti-globalism

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and extremism, and the depression, anxiety, and schizophrenia that have forced themselves into the center of public attention are its dark side. Every new and every retrograde change in the cultural context points to one common underlying economic condition: recession. Behind all of these cultural moods lies a shared anxiety about a single question: when life will no longer get better, how are we to go on living?

Since the third quarter of 2022, the Chinese economy has displayed enduring recessionary features: persistently high unemployment, weak consumption, depressed price levels, and sluggish investment. Figure 1 directly shows the trends of three key indicators. The consumer price index (CPI) fell sharply in the second half of 2022 and dropped below zero in the third quarter of 2023; the producer price index (PPI) and the purchasing managers' index (PMI) had already shown sustained downward trends as early as 2021 and have fluctuated around zero since 2022. The CPI reflects the price level of the final products purchased by consumers, the PPI reflects price changes at the production stage, and the PMI is an expectations-based indicator of industrial activity. Taken together, these three indicators show that the Chinese economy is undergoing a contraction that runs through the entire supply chain.

Although the PPI and PMI had begun to decline markedly in 2021, signaling cooling supply-side activity, this was because pandemic-control measures had restricted industrial production, and the CPI was at the time rising significantly because supply was constrained; at the indicator level, therefore, this did not constitute deflation. In July 2022, pandemic controls were fully lifted across the country. Industrial production, which had been gradually recovering, restarted completely, yet global market demand was weak. A large number of manufacturers had no choice but to "switch from exports to domestic sales." Combined with the impact of the pandemic on household income, the CPI began to fall sharply. The third quarter of 2022 can therefore, at the least, be regarded as the onset of the current deflation. The fear of deflation so often invoked during the lockdown era has finally materialized.

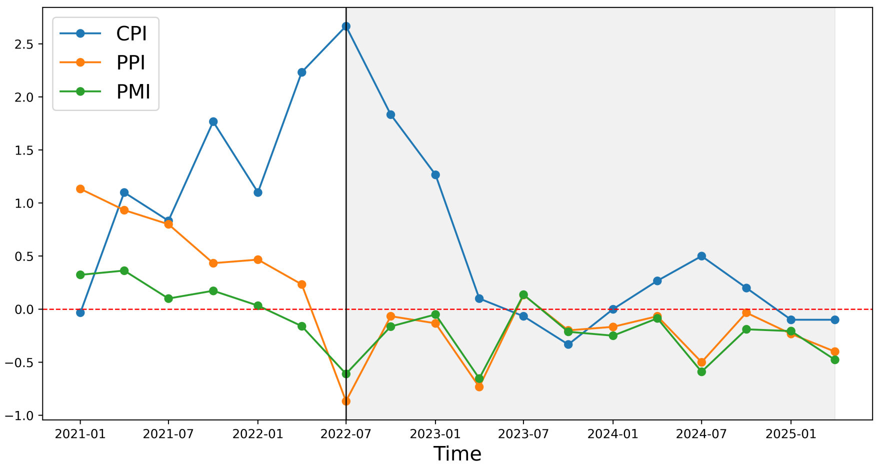


Figure 1: The main manifestations of deflation: PPI, CPI, and PMI.
Data source: National Bureau of Statistics.

Looking at the indicators of the first quarter of 2021, the CPI was already below zero, which at least suggests that China's demand-side consumption capacity was already in trouble before the pandemic shock. International demand kept the PPI and PMI in positive territory, masking, to some extent, the risk of a weak domestic market. We should therefore at least suspect that insufficient domestic consumption is the deep cause of the current deflation, with the pandemic shock as the catalyst. This implies that the current deflation is caused by structural problems rather than being a one-off event driven by an exogenous shock.

1.2 Impacts

Deflation refers to a sustained decline in aggregate demand, aggregate supply, and the price level, and is generally regarded as an unhealthy economic state. When an economy falls into deflation, every component of the national income identity is affected, generating a self-aggravating cycle:

$$GDP = C + I + G + (X - M)$$

On the supply side, small and medium-sized enterprises act as price takers in perfectly competitive markets, and their factor-input decisions follow the first-order condition that the marginal product equals the real wage. When a falling price level raises the real wage, a firm must raise the marginal product of each individual worker to maintain zero profits. With working hours fixed, the only way to raise the marginal product is to reduce labor input, which inevitably leads to layoffs.

On the demand side, persistently falling prices generate deflationary expectations among consumers and a “buy on the rise, sell on the fall” psychology, delaying consumption (C) and investment (I). This delay further suppresses aggregate demand and accelerates the downward drift of prices. At the same time, falling prices and shrinking demand compress firms’ profit margins, pushing them toward cost control or output reduction. Large firms are more inclined to lay off workers, while small and medium-sized enterprises face a higher risk of shutdown. Layoffs and bankruptcies push up the unemployment rate and reduce household income, forming a vicious cycle: falling income, reduced consumption, shrinking demand.

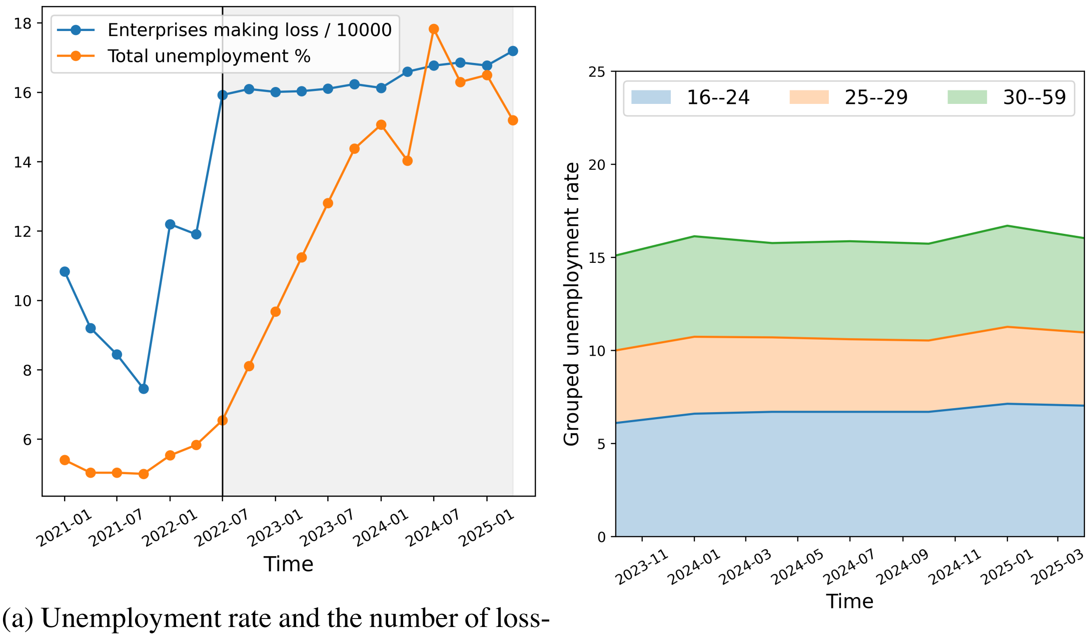
For the government, household unemployment and income losses directly reduce income tax revenue and, through the consumption contraction, indirectly reduce consumption tax revenue. At the same time, fiscal expenditures such as unemployment benefits, social insurance, enterprise support, and stimulus programs[1] rise substantially. This imbalance between revenue and expenditure deteriorates the fiscal position and forces the government to cut public spending (G), further depressing aggregate demand.

In an era of deflation, the biggest shock facing households is unemployment, and the biggest shock facing firms is losses. Figure 2a shows the trends in China’s total unemployment rate and the number of loss-making firms since 2021. After deflation began, the number of loss-making firms jumped quickly to a high of about 160,000 and then drifted upward to about 170,000; the unemployment rate climbed rapidly from around 7 percent, peaked at close to 18 percent, and was barely pulled back to around 16 percent by a series of employment-promotion policies.

Figure 2b breaks down the stacked changes in unemployment by age group, where the vertical width of each band represents the relative unemployment rate. The 16–24 age group has the highest unemployment rate, followed by the 30–59 group, while the 25–29 group is relatively lower. The 25–29 group includes graduate students and other highly educated workers as well as people with relevant work experience, and it has not yet crossed the threshold of workplace age discrimination, so its employment

[1] All figures are produced with matplotlib (Hunter, 2007).

rate is relatively high. The 16–24 group, by contrast, covers most college graduates as well as some 16–18-year-olds without higher education; they lack work experience and still need to explore career options. Taken together, these two observations show that China's current labor market is extremely poor, and especially unfavorable to young people newly entering it. The 30–59 group is also informative. If the employment rate of graduates reflects the expansion of the labor market, the unemployment rate of the middle-aged group reflects its contraction. Persistently high unemployment among the 30–59 group shows that China's labor market is not merely failing to expand; it is contracting. Moreover, middle-aged workers are usually the breadwinners of their families, and their unemployment hits household finances and living standards harder.



(a) Unemployment rate and the number of loss-making firms.

(b) Unemployment rate by age group.

Figure 2: Unemployment and loss-making firms by quarter.

Data source: National Bureau of Statistics.

More dangerously, deflation carries the potential to damage the long-run growth mechanism of the economy. According to the *Growth Report: Strategies for Sustained Growth and Inclusive Development* published by the World Bank in 2008 (Commission on Growth, 2008), economies with sustained long-run growth share five common features: openness, macroeconomic stability, high saving and investment rates, market allocation, and appropriate leadership and governance. The long-run risks of deflation are:

1. **Reduced openness.** When international demand is weak, absorbing capacity through domestic consumption alone may require restricting imports.

2. **Impaired macroeconomic stability.** Combating deflation may require unconventional monetary policy and frequent fiscal stimulus, which can destabilize the macroeconomy.

3. **Lower saving and investment rates.** Shrinking household income lowers the saving rate, and without profitable investment opportunities no new investment is generated.

4. **Reduced efficiency of market allocation.** An economy-wide contraction easily leads to market failure, and deflation management often requires stronger government intervention, which distorts market allocation.

5. **Deteriorating governance.** The fiscal pressure caused by deflation can tightly constrain governance and routine administration, weakening the capacity and efficiency of the government.

In sum, deflation is a self-reinforcing vicious cycle that forms a complex feedback mechanism spanning production, consumption, investment, and government spending. Rising unemployment and corporate losses not only directly affect household income and consumption capacity but also create extremely serious social problems. In the long run, deflation may destroy the economy's basic growth mechanism and lead to deeper structural problems. Deflation is a cycle because, starting from any sector, one can trace a chain of causation that runs through the whole economy and loops back on itself. Where, then, is the source of deflation? Which sector first set it in motion?

2 Mechanisms

Deflation accompanying economic recession is often labeled "malignant" deflation. A recent study by Xia (2023) compiles the three most famous episodes of "harmful deflation" in world economic history, as shown in Table 1.

Table 1: Cases of harmful deflation compared

	The U.S. Great Depression of the 1930s	Japan's "Lost Decade" of the 1990s	The 2008 Global Financial Crisis
Primary causes	The Fed's contractionary monetary policy to control the Wall Street bubble triggered a sharp fall in aggregate demand	Asset bubble burst, weak domestic demand, overcapacity, slow industrial restructuring	Real estate market collapse, credit market contraction, global financial turmoil
Policy responses	Contractionary policies	Low interest rates, QE, fiscal stimulus programs	Zero interest rates, multiple rounds of QE, economic stimulus programs
Outcomes	Misguided responses led to a deflationary spiral and severe recession	Long-standing structural problems limited the effectiveness of the responses; recovery stayed weak	Effectively stabilized financial markets, avoiding a Great-Depression-style deflationary spiral and recession

Data source: Xia Guangtao, "The intellectual origins, basic concepts, and historical experience of deflation theory: A theoretical review," 2023, *Cai Jing Zhi Ku* (Finance Think Tank).

The bursting of financial bubbles appears to be an important cause of major deflationary episodes. During the formation and inflation of a financial bubble, real asset prices are overvalued and the demand for liquidity in money markets rises sharply, inducing the monetary authority to issue more money. Money growth actually far outruns the growth of asset values, and the market begins trading fictitious symbols detached from value. Once a financial scandal or some other event pricks the bubble, people realize the true value of the assets they hold; panic triggers selling, and asset prices go into free fall. The assets held by banks and other financial institutions shrink in value, balance sheets become unbalanced, and some debtors lose their ability to repay, creating bad debts. Bad debts further write down creditors' assets, and clients begin to question the institutions' financial risk, which can trigger bank runs alongside the mass selling. Funds flow out of financial markets in large volumes, causing a shortage of financial supply that constrains corporate investment; meanwhile, the real money depositors withdraw is less than the nominal money they put in during the bubble (Zhu, 2019), and out of fear and pessimism about financial crises, households tend to cut consumption. That is, consumption falls while savings fail to become capital, which leads to a broad-based, sustained decline in the price level as manufacturing shrinks back to the economy's real scale. This is deflation.

Asset bubbles do not appear without warning either. The simplest summary of a financial bubble is that the fictitious economy expands faster than the real economy, so that the additional growth is fictitious value sustained purely by market confidence and lacking real-asset backing. A classic example is the 2007 subprime crisis in U.S. asset-backed securities. To counter the market downturn caused by the bursting of the 2001 Internet bubble and the September 11 attacks, the Fed cut interest rates sharply and kept them low for a long time to stimulate the market. This gave the capital-intensive real estate industry access to abundant cheap loans, and the industry expanded dramatically. At the same time, banks, flush with capital, kept lowering the bar for personal loans and launched various complex new loan products, greatly stimulating housing demand. With supply and demand in place, the U.S. housing market grew rapidly. Although rigid housing demand was soon satisfied, the loose monetary policy was not withdrawn; the continued rise in house prices created the illusion that housing was a low-risk, high-quality asset, and speculative demand intensified. Financial markets then began issuing securities backed by mortgage loans, namely mortgage-backed securities (MBS), spreading the risk of real estate assets across the entire financial market. As speculative demand for housing gradually became saturated, and the Fed raised rates modestly in 2006, U.S. house prices stopped rising and soon began to fall. The market quickly recognized the bubble nature of housing. Banks re-evaluated subprime loans and discovered massive bad debts; balance-sheet imbalances drove several banks into bankruptcy. MBS depreciated sharply, bankrupting the many financial institutions that held them. Household-sector assets shrank, capital shifted to safe havens, and consumption was cut…

Excess money issuance and asset bubbles are two sides of the same coin: excess issuance is an important driver of bubbles and prepares the ground for post-bubble deflation. Tang and Zhang (2020) summarize two mechanisms. According to the money-decomposition theory, excess money issuance raises prices, but irrational rises in asset prices absorb the excess money, keeping it from flowing into the real economy and creating a crowding-out effect; households and firms face higher prices and costs with unchanged incomes and therefore cut consumption and investment, steering the economy toward deflation. According to the asset-equilibrium theory, if the rate of return in the fictitious economy exceeds that in the real economy, the excess money flows into the fictitious economy and creates a state of excess demand there, so money destined for consumption also flows out of the real economy, crowding out consumption and restricting real economic development, sowing the seeds of deflation. The same study also estimates that from 2008 to 2018, the liabilities of Chinese real estate firms directly accounted for about one-third of new aggregate financing; counting households and non-credit sectors that mainly purchase housing with credit funds, the share rises to two-thirds. This shows how China's real estate market has eroded the real economy.

In short, the disorderly expansion of financial markets can largely be regarded as a precondition for major deflationary episodes in economic history, and the expanding financial sector is usually real estate. To analyze the rapid expansion of China's real estate market, one inevitably must discuss the development path.

3 China's Development Path

3.1 The Demographic Dividend and the World Factory

Latecomer countries have few options for achieving economic growth, let alone catching up with the developed world. Scarce capital, backward technology, peripheral position, colonial history — all of these severely constrain the development paths available to developing countries. The common paths of developing countries start from their own resource endowments: oil-rich countries export oil, such as Iran, Saudi Arabia, and the United Arab Emirates; countries rich in natural resources export agricultural products and minerals, such as the Congo, Thailand, Argentina, and Chile. If natural resources are also scarce, or if a country for strategic reasons does not want to rely heavily on mineral exports, and it has a large low-income population, it can adopt an "import substitution" strategy.

Import substitution means focusing on the industries with heavy imports, using protective and industrial-support policies to gradually build up domestic production, replace imported products, and finally turn to exports. In the ideal case, import substitution quickly accumulates capital and technology, builds a domestic industrial system in a relatively short time, and achieves economic self-reliance and rapid growth. But the initial endowments of developing countries dictate that import substitution must begin with basic low-end industrial processing; moreover, international trade requires comparative advantage, which means developing countries must keep manufacturing costs low over the long run.

3.1.1 A Stunted Middle Class

At the start of reform and opening up, China's industry was still relatively backward, although earlier development and Soviet aid had laid a certain foundation, and its technological level was roughly at the level of low-end processing. In terms of comparative advantage, a large low-income population supplied manufacturing with abundant cheap labor, giving labor-intensive low-end industries a labor-cost advantage, and the sheer size of the population also gave foreign capital the comparative advantage of proximity to the market. With this "demographic dividend" and a series of policies epitomized by "trading the market for technology," China rapidly built a complete industrial system after reform and opening up and turned from an industrial importer into an exporter.

This process required keeping wage levels relatively stable: once wages rise, labor costs rise, China's comparative advantage in labor-intensive manufacturing weakens or is lost, and because the marginal propensity to consume of wage earners is lower, the additional consumption cannot fill the demand lost to higher export prices. Moreover, the development of high-end industries requires large capital investment and presupposes well-developed modern infrastructure, and the cultivation of human capital requires education investment on an enormous scale…It was therefore necessary to maintain, or even raise, manufacturing profits, keep workers' wages stable, compress social security, and maintain a high saving rate, so as to fund the push into high-end industries. At the time, China's GDP growth came mainly from exports, investment, and government spending, so constraining household consumption had no major consequences. So in China at that time, an appropriately tilted distribution helped capital accumulation, and inequality became the "necessary evil" of economic development. As a result, China never managed to cultivate a strong middle class the way the United States and Japan did.

3.1.2 Overcapacity

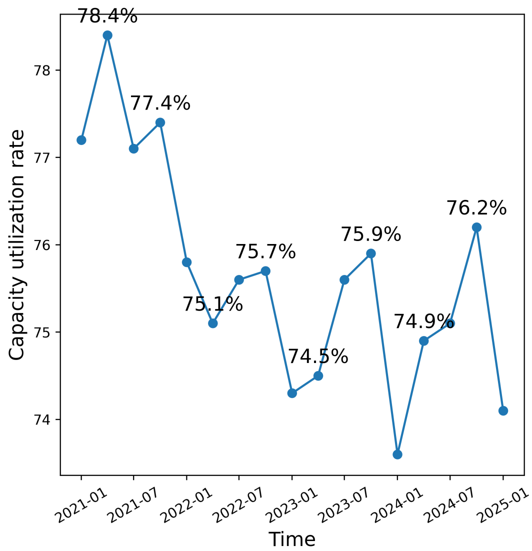
Another hidden danger that the import substitution strategy planted in the Chinese economy is overcapacity. Import substitution aims to accumulate capital and technology through international trade, develop domestic production to replace imports, and turn to exports — a process that requires strict wage control. The industrial scale of an economy pursuing import substitution is always benchmarked to the global market. After China developed its full industrial system through import substitution, why should it stop? Why not expand into the global market? The preserved demographic dividend and other factor endowments could earn large surpluses once exported. And so we became the "world's factory": a super-sized trade surplus nourished several world-leading high-tech industries and a huge wealthy class, while workers' incomes and treatment were held low for a long time. This continued until, in the words of a popular saying, "one pot of steel from a Chinese mill could meet the entire world's demand for ballpoint pen tips"[2] — and then other industrial countries began to accuse China of "overcapacity" and of tolerating dumping.

When the world market approaches saturation, Chinese firms going abroad display their most practiced skills: price wars and supply-chain competition. The former rests on China's ability to maintain price competitiveness countercyclically on the basis of low wages and low welfare; the latter on China's possession of the full industrial chain and super-sized capacity, able to meet almost any large (and sometimes any small) demand within a short time. But when the world economy cools, or other reasons shrink international demand, the super-sized capacity that China tailored to world demand falls into surplus. For individual economic agents, this means large-scale suspension of work, closure, layoffs, and bankruptcy.

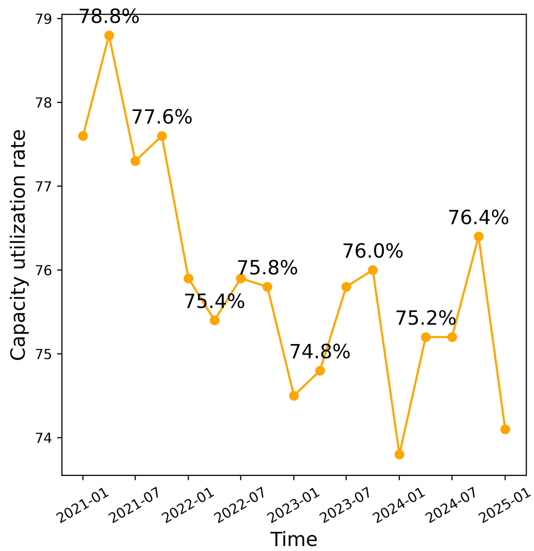
Figure 3a shows the quarterly fluctuation of China's overall capacity utilization over the past four years, and Figure 3b shows the fluctuation of manufacturing capacity utilization. Well before the pandemic, capacity utilization had already begun to fluctuate downward, and after the pandemic shock it never recovered; the trough once approached

[2] This claim may be apocryphal; it is quoted here only for rhetorical effect.

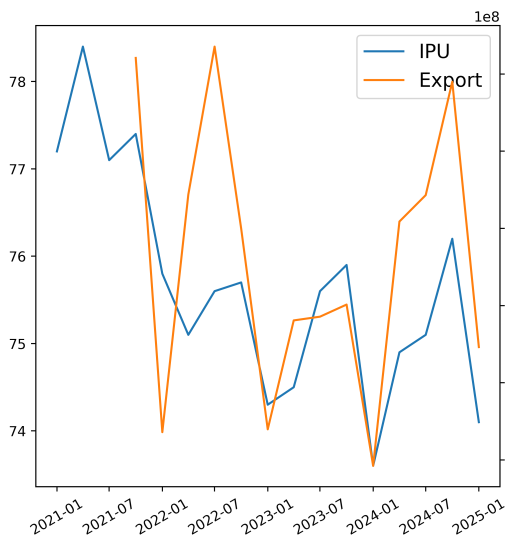
73 percent, meaning that jobs corresponding to about 27 percent of total capacity could not operate normally. Manufacturing capacity utilization was slightly higher than the overall figure in most periods, indicating that the non-manufacturing sector cooled more severely. More importantly, Figures 3c and 3d show that the fluctuation of China's capacity utilization is highly correlated with the fluctuation of export values, confirming the overcapacity hazard of China's role as the world's factory.



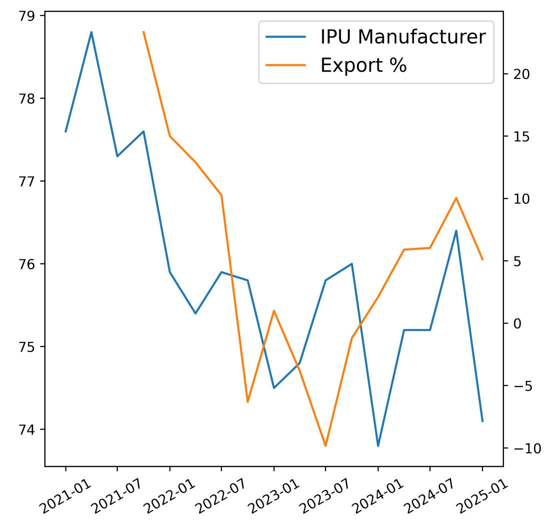
(a) Overall capacity utilization.



(b) Manufacturing capacity utilization.



(c) Capacity utilization and exports (averages).



(d) Manufacturing capacity and export growth rates.

Figure 3: Capacity utilization and export values by quarter.

Data source: National Bureau of Statistics.

3.2 The Tax-Sharing Reform and Land Finance

In 1994, China implemented the tax-sharing reform, readjusting the distribution of tax revenue between the central and local governments. Its main purposes were to expand the central government's fiscal resources and strengthen its macroeconomic-control capacity and, in line with building the socialist market economy, to strengthen the consistency of the domestic market — a lineage that continues in the current push for a "unified national market" aimed at weakening local protectionism. An important logical problem is that the tax-sharing reform moved taxing power upward without moving spending responsibilities upward, and subsequent policy adjustments delegated more concrete spending responsibilities down to local governments. This mismatch between fiscal power and spending responsibilities often left local governments in fiscal straits, creating the incentive to seek revenue "outside the system." The model of relying on land-transfer fees for fiscal revenue, making the fiscal position highly dependent on land sales, is called "land finance."

Land is the most important asset local governments actually own; in regions lacking industry and natural resources, land transfers basically become the only source of local revenue. But barren land itself does not command a high price. To obtain more funds as collateral from banks and to negotiate higher transfer fees with real estate developers, local governments must find ways to push land prices up. Strict regulation and rational control of the real estate industry become almost impossible under such circumstances. The business model of local governments is broadly uniform: transfer land parcels to developers and promise to bias new public service provision toward those parcels. The most concrete example is that the moment buildings start rising in a "new district" on the urban fringe, rumors immediately circulate about planned hospitals, schools, parks, shopping malls, and high-grade roads nearby — some real, some not. Moreover, to further raise and maintain property prices, local governments have every incentive to encourage rural residents to move to cities, gradually tighten the approval of rural homesteads, plan as little infrastructure in townships as possible, and impose a series of residence-based restrictions on school enrollment, medical care, and social insurance participation, circuitously forcing farmers and migrant workers to buy housing. Finally, because local governments have formed a community of interest with banks and real estate developers within this mechanism, they have no reason (and sometimes no ability) to ask banks to strengthen loan review and risk supervision, or developers to control leverage risk; the result is the indulgence of enormous financial risk. For the government itself, once hooked on the seductive cycle of selling land, inflating house prices, collecting revenue, and winning promotion, local governments find it easier and more natural to borrow heavily through urban investment companies and build vanity projects.

The harms of this model are, first, that it greatly distorts the rational allocation of public service provision, in both time and space; second, that it indulges, shields, and even actively abets the unbridled expansion of the real estate market; and third, that it "empties the six pockets" — the young couple and both sets of parents — crowding out consumption in the real economy. As housing and land prices rise, banks hold large amounts of real estate as collateral, households rely heavily on credit to buy homes, and government finances depend heavily on land prices — a large part of the entire financial market rests on the inflated fictitious value of real estate. Distribution and consumption were deliberately ignored; history has repeated itself, and crisis is imminent.

3.3 Real Estate and Local Government Implicit Debt

The three pieces of the crisis puzzle seem complete: a shrinking consuming class (insufficient domestic demand), surplus production capacity (unemployment), and a financial bubble (write-downs and bankruptcies). At this point, local government implicit debt acts as an amplifier, magnifying the breadth and intensity of financial risk. A machine-learning-based empirical study finds that debt is the most important factor in producer-sector deflation, accounting for about 38 percent of the contribution (Chen et al., 2021).

Local government implicit debt and the disorderly expansion of the real estate market go hand in hand. First, the government must spend money to honor its promises of higher land prices: building infrastructure and commercial complexes requires large capital outlays. Second, the expectation of future high land revenues obscures the government's debt-service pressure. Third, the GDP-based evaluation of officials' performance naturally tends to produce non-intertemporally-optimal behavior. Finally, the oversized real estate market ties up government funds, making it hard to meet the funding needs of routine governance, social security, urban renewal, and construction. Given all of these factors, local governments seem compelled to expand debt financing; deficits on the balance sheet are monitored by higher-level governments, so implicit debt through platforms such as urban investment companies becomes the natural choice.

Implicit debt is not directly booked in the government's revenue and expenditure budget. It is usually borrowed in the name of companies, but with credit backing such as government guarantees it obtains higher amounts, more flexible maturities, and more favorable interest rates. Implicit debt is hard to track and control, its funding sources are scattered and broad, and its credit risk is often underestimated. The result is that financial risk transmits rapidly to every corner of the economic system; and when crisis arrives and local governments plan to expand government purchases to stimulate demand, they face debt constraints. Moreover, because the central government in effect serves as the ultimate debtor, its use of expansionary monetary and fiscal policy is likewise constrained by implicit debt. In summary, implicit debt amplifies and transmits financial risk, and limits the capacity to respond to crisis.

3.4 Distributional Imbalance and Insufficient Demand

If China's development path has a single main thread, it is the neglect of balance in the distribution structure.

Receiving the relocation of low-end manufacturing required tight control of labor costs, and developing high-end manufacturing required capital accumulation, so the household sector long maintained a state of low wages and a high saving rate. Moreover, because the development of China's industrial system depended little on domestic demand, there was little incentive to boost domestic consumption. The disorderly expansion of land finance and real estate directly drained household-sector liquidity and, by overdrawing credit, constrained new household consumption over the long run, while linking quality of life to homeownership widened the urban–rural gap further. Local government implicit debt put pressure on social governance and social security: in June 2025, for example, the National Audit Office disclosed RMB 60.161 billion in irregularities in the urban and rural pension insurance funds, of which RMB 40.626 billion had been diverted to "three guarantees" expenditures and government debt repayment, among other uses.[3] Low social welfare, together with insufficient, unequal, and insecure social security, makes residents prefer to cut current consumption in order to maintain high precautionary savings. In addition, living standards and welfare that go

[3] National Audit Office of the People's Republic of China, *Report on the Audit of the Implementation of the Central Budget and Other Fiscal Revenue and Expenditure in 2024*, June 24, 2025.

unimproved for long periods have greatly lowered the total fertility rate; combined with population aging, aggregate demand shrinks further, with long-run consequences.

In sum, China's development path led almost inevitably to distributional imbalance, which in turn produced insufficient domestic demand and, together with the real estate financial bubble, formed a "deflationary spiral." This also means that China's current deflationary crisis is a structural problem that can only be resolved through systemic reform.

The discussion so far has argued, at the descriptive level, that the insufficient demand caused by distributional imbalance is the underlying cause of China's current deflation. The next step is to argue and test this with the help of an economic model.

4 The Economic Model

To represent the distribution structure intuitively in an economic model, one must distinguish middle- and low-income groups from high-income groups by some criterion. Theoretical models usually involve no specific numbers, so a direct income-based division is not possible. One possible approach is to distinguish the groups by their idiosyncratic properties: for example, high-income groups have a lower marginal propensity to consume while middle- and low-income groups have a higher one; or, the lower a group's average income, the more its income composition relies on wage income from selling labor, whereas high-income groups enjoy more capital income. Moreover, capital income is naturally more concentrated while labor income is more dispersed; if the capital income share in an economy is too high, it is likely to have excessively high wealth concentration as well, and distribution becomes more unequal. This essay therefore divides income into "labor income" and "capital income," whose structure at least partly reflects how the economy's output is distributed.

4.1 Model Setup

Consider a closed three-sector, two-factor economy at full employment. Total income is divided into "labor income" and "capital income." The government taxes labor and capital income at different rates and adjusts the labor–capital income ratio through transfers. To simplify the analysis, we assume no adjustment frictions and no transfer costs.

Labor and capital income. Economic income consists entirely of labor income and capital income, with the labor income share equal to α. Labor income comes from the wage (W) of labor (N); this part is taxed at rate t_L, and it also includes untaxed transfers (Tr). Capital income comes from capital returns (RK) and is taxed at rate t_K. Disposable labor and capital income are therefore:

$$Y_L^d = (1 - t_L)\alpha\bar{Y} + \frac{Tr}{P} = \frac{(1 - t_L)WN + Tr}{P} \tag{1}$$

$$Y_K^d = (1 - t_K)(1 - \alpha)\bar{Y} = \frac{(1 - t_K)RK}{P} \tag{2}$$

Government budget constraint. Government spending includes government purchases and transfers. Since government purchases are not the focus of the discussion, they are set to a constant G_0; all government revenue comes from taxes on labor and capital, as in Equation 3:

$$\begin{aligned} PG_0 + Tr &= t_L WN + t_K RK = t_L \alpha P\bar{Y} + t_K(1-\alpha)P\bar{Y} \\ G_0 + \frac{Tr}{P} &= \left[t_L\alpha + t_K(1-\alpha)\right]\bar{Y} \end{aligned} \tag{3}$$

Total consumption function. In the whole economy, total consumption is composed of the consumption of the labor and capital sectors, which have different marginal propensities to consume; following the classical microeconomic assumption, we take $\beta_L > \beta_K$. Combining Equations 1 and 2 gives Equation 4:

$$\begin{aligned} C &= \beta_L Y_L^d + \beta_K Y_K^d \\ &= \beta_L\left[(1-t_L)\alpha\bar{Y} + \frac{Tr}{P}\right] + \beta_K\left[(1-t_K)(1-\alpha)\bar{Y}\right] \end{aligned} \tag{4}$$

Aggregate demand. Aggregate demand consists of total consumption plus total investment and total government spending. To simplify the analysis, investment and government spending are not considered and are set to constants:

$$AD : Y^d = C + I_0 + G_0 \tag{5}$$

Quantity equation. The money market obeys the Fisher equation. Since $\beta_L > \beta_K$, when the labor income share α rises, more money enters circulation through consumption, so the velocity of money V is a function of α with a positive first derivative:

$$MV(\alpha) = P\bar{Y}, \quad \frac{\partial V}{\partial \alpha} > 0 \tag{6}$$

Long-run equilibrium. Total output equals aggregate demand:

$$\bar{Y} = C + I_0 + G_0 \tag{7}$$

4.2 Policy Scenarios

We now use the model to analyze policy scenarios. Because monetary policy has no direct effect on the income distribution structure, is rendered powerless by the zero lower bound in a deflationary environment, and is regarded by rational expectations theory as ineffective in the long run, this subsection focuses on fiscal policy.

4.2.1 Fiscal Neutrality

When fiscal policy does not actively adjust the income distribution, a distribution structure tilted toward labor for other reasons breaks the government budget balance. According to the government budget constraint in Equation 3, the government must adjust the price level P to restore balance, so the change in consumption demand is an increase in labor consumption minus a decrease in capital consumption:

$$\Delta C = \beta_L\left[(1-t_L)\Delta\alpha\bar{Y} - \frac{Tr}{P^2}\Delta P\right] - \beta_K\left[(1-t_K)\Delta\alpha\bar{Y}\right] \tag{8}$$

Differentiating the above expression gives:

$$dC = \beta_L \left[(1-t_L)\bar{Y}d\alpha - \frac{Tr}{P^2}dP \right] - \beta_K(1-t_K)\bar{Y}d\alpha = 0 \tag{9}$$

$$\beta_L \frac{TR}{P^2}dP = \bar{Y}d\alpha \left[\beta_L(1-t_L) - \beta_K(1-t_K) \right] \tag{10}$$

Now let $\Lambda \equiv \beta_L(1-t_L) - \beta_K(1-t_K)$. In China's actual economy, the tax rate faced by rank-and-file workers is far lower than that faced by capital holders, so $\Lambda > 0$, and hence:

$$dP = \frac{\Lambda P^2 \bar{Y}}{\beta_L TR}d\alpha > 0, \quad \frac{dP}{d\alpha} = \frac{\Lambda \bar{Y}}{\beta_L Tr}P^2 > 0 \tag{11}$$

Thus $P'(\alpha) > 0$, meaning that the price level increases with the labor income share.

4.2.2 Active Fiscal Policy

Suppose the government extracts capital income to fund transfers subsidizing workers. In this process α increases, meaning the labor income share expands and the capital income share shrinks; the labor tax rate stays unchanged while the capital tax rate rises. We then have:

$$\begin{aligned} d(\frac{Tr}{P}) &= \underbrace{(t_L \bar{Y} d\alpha)}_{\text{higher labor tax}} - \underbrace{(t_K \bar{Y} d\alpha)}_{\text{lower capital tax}} + \underbrace{(1-\alpha)\bar{Y}dt_K}_{\text{revenue from higher capital tax}} \\ &= (t_L - t_K)\bar{Y}d\alpha + (1-\alpha)\bar{Y}dt_K \end{aligned} \tag{12}$$

Because we assume that share adjustment is costless, the entire rise in the labor share can be attributed to a higher capital tax rate, so $dt_K = \phi d\alpha$ with $\phi > 0$. Let $\kappa d\alpha = d(Tr/P)$. The change in total consumption can then be expressed as:

$$\begin{aligned} dC &= \beta_L \left[(1-t_L)\bar{Y}d\alpha + \kappa d\alpha \right] - \beta_K \left[(1-t_K)\bar{Y}d\alpha + (1-\alpha)\bar{Y}\phi d\alpha \right] \\ &= d\alpha \{ \beta_L \left[(1-t_L)\bar{Y} + \kappa \right] + \beta_K \left[-(1-t_K)\bar{Y} - (1-\alpha)\bar{Y}\phi \right] \} \end{aligned} \tag{13}$$

This expression reveals the overall effect of redistribution policy on total consumption. On the one hand, a higher labor income share and larger transfers raise workers' disposable income and push consumption up; on the other hand, lower capital income and a heavier tax burden may suppress the consumption of capital holders. When workers' marginal propensity to consume is high, that is, $\beta_L > \beta_K$, and ϕ is not too extreme, redistribution on balance tends to expand total consumption.

4.2.3 The General Dynamic Equation

Combining the monetary and fiscal factors, the dynamic equation of the price level is:

$$P = \frac{MV(\alpha)}{\bar{Y}} \times \exp \left(\int_0^\alpha \underbrace{\frac{\beta_L \left[(1-t_L) - \theta(\xi) \right] - \beta_K \left[(1-t_K) - \psi(\xi) \right]}{\beta_L Tr/(P^2 \bar{Y})}}_{\text{adjustment kernel}} d\xi \right) \tag{14}$$

Here, $\theta(\xi)$ describes the transfer rule of fiscal policy on the workers' side, and $\psi(\xi)$ describes the rule for taxing and transferring capital income. The numerator $\beta_L[(1-$

$t_L) - \theta(\xi)] - \beta_K[(1-t_K) - \psi(\xi)]$ reflects the marginal-consumption shift of the different groups under after-tax income and fiscal subsidies; the denominator $\beta_L Tr/(P^2\bar{Y})$ represents the capacity of fiscal transfers to amplify or absorb price changes. Regardless of the fiscal policy, as long as $\beta_L > \beta_K$ and $V'(\alpha) > 0$, raising α eventually raises P, though fiscal policy can change the magnitude.

4.2.4 Summary

The theoretical derivation in this section uses labor income to proxy for the middle- and low-income groups that in reality rely mainly on wage income, and capital returns to proxy for high-income groups, and demonstrates at a mathematical level that the distribution structure is linked to the price level. When redistribution and similar measures raise the labor income share and move the distribution structure toward balance, the price level shows a clear upward trend; when the distribution structure tilts toward capital for a long time without timely correction through fiscal instruments, the structural shortfall of consumption demand depresses the price level and generates persistent deflationary pressure. We can therefore state the following hypotheses:

- **H1:** Other things equal, an unbalanced distribution structure lowers the price level.
- **H2:** Other things equal, an improved distribution structure raises the price level.

5 Empirical Analysis

To test the hypotheses above, this section builds a reduced-form estimation model and runs regressions on cross-country data for 77 economies from 2011 to 2021. Because different economies have idiosyncratic features that may affect the effect of the distribution structure on the CPI, we use fixed-effects ordinary least squares (FE-OLS), controlling for individual fixed effects. To avoid the endogeneity problem of two-way causality, the CPI is led by one period. The final model specification is:

$$CPI_{i,t+1} = Constant + \beta_1 Distribution_{it} + \sum_{k=1}^{k} \theta_k X_{it}^k + \delta_i + \epsilon_{it}$$

Here, $CPI_{i,t+1}$ is the consumer price index of economy i in year $t+1$; $Constant$ is the intercept; $Distribution_{it}$ is the distribution-structure indicator — to avoid estimation bias, we use both the Gini coefficient and the income share of the bottom 20 percent — and its coefficient β_1 is the coefficient of interest, reflecting the effect of the distribution structure on deflation; θ_k is the coefficient on the k-th control variable; X_{it}^k is a set of control variables; δ_i is the individual fixed effect, which absorbs unobservable idiosyncratic characteristics that affect the CPI; and ϵ_{it} is the error term.

The control variables include the central government debt ratio, the employment rate of the population aged 15 and above, the domestic private-sector credit ratio, social security contributions, and the labor force participation rate. The central government debt ratio represents the degree of fiscal expansion, which affects the CPI directly through spending expansion and monetized financing; the employment rate is related to economic vitality and household income, affecting the price level through investment and consumption; the private-sector credit ratio reflects the financing environment and the degree of demand expansion; and social security spending may affect prices indirectly through the marginal propensity to consume. Controlling for these variables yields cleaner estimates of the effect.

Table 2 reports the definitions and descriptive statistics of the variables used in this section. To avoid distorting the data and to ensure the realism of the estimates, the variables are not transformed. All data come from the World Bank.

Table 2: Descriptive statistics

Variable	Description	N	Mean	Std. dev.	Min	Max
CPI	Consumer price index (2010 = 0)	784	24.41	43.08	-2.255	436.5
gini	Gini coefficient	784	35.66	7.670	23.20	63.40
income_low20	Income share of the bottom 20% / GDP	784	7.004	1.819	2.400	10.50
gov_debt_gdp	Central government debt / GDP	784	60.13	32.64	6.160	212.4
emp_r	Employment rate, population aged 15+	784	56.68	7.946	34.56	85.84
credit_priv_gdp	Domestic private-sector credit / GDP	784	105.0	80.58	10.12	407.4
social_contrib	Social security contributions	784	22.75	13.63	0.006	58.09
labor_part_r	Labor force participation rate	784	61.25	6.607	39.50	86.49

Table 3 reports the regression results. In the regression for next-period CPI, the current Gini coefficient enters negatively and is significant at the 95 percent level or better, indicating that the more unequal the income distribution, the lower the price level is likely to be — **H1** is confirmed. The coefficient on the income share of the bottom 20 percent is significantly positive, indicating that improving the income of low-income groups helps raise future price levels — **H2** is confirmed. Following Guiso et al. (2015), we also measure the economic significance of the distribution coefficient: $ES_x = |bs_x/s_y|$. The results show that a one-standard-deviation increase in the Gini coefficient lowers the CPI by 0.624 standard deviations, and a one-standard-deviation increase in the income share of the bottom 20 percent raises the CPI by 0.621 standard deviations. The higher the Gini coefficient and the lower the bottom-20 percent income share, the more unequal the income distribution. Taken together, the regression results show that balancing the income distribution structure can effectively raise the price level, helping to avoid and resolve deflation. The two hypotheses thus hold in economic terms.

Table 3: FE-OLS estimation results

	F.CPI			
	(1)	(2)	(3)	(4)
gini	-4.032***	-3.091**		
	(1.301)[0.812]	(1.235) [0.624]		
income_low20			14.377**	13.315**
			(5.947) [0.700]	(6.186) [0.621]
gov_debt_gdp		0.772***		0.784***
		(0.174)		(0.169)
Controls	N	Y	N	Y
FE	Y	Y	Y	Y
R^2	0.063	0.207	0.058	0.214
N	698	698	698	698

Notes: Cluster-robust standard errors in parentheses; economic significance in brackets. $^{*}\ p < 0.1$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

6 Policy Simulation

6.1 Model Construction

This section builds a three-sector New Keynesian dynamic stochastic general equilibrium (DSGE) model of a closed economy with heterogeneous households. Households are divided into middle-/low-income and high-income types. The government taxes their income at different rates and controls the shares of the two types by adjusting tax rates and transfers.

6.1.1 Middle- and Low-Income Households

The middle- and low-income household supplies labor N^L and capital K^L. Its objective is to maximize the discounted present value of lifetime utility (15), subject to the budget constraint (16); the Lagrange method yields the optimal decision problem under constraints, namely the Euler equation (17); the first-order condition for labor supply is (18); and capital accumulation satisfies (19).

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\ln C_t^L - \frac{(N_t^L)^{1+\phi}}{1+\phi} \right) \tag{15}$$

$$C_t^L + I_t^L + \delta K_t^L = (1-\tau_L) w_t N_t^L + r_t K_t^L + Tr_t \tag{16}$$

$$\frac{1}{C_t^L} = \beta \mathbb{E}_t \left[\frac{1}{C_{t+1}^L} (r_{t+1} + 1 - \delta) \right] \tag{17}$$

$$(N_t^L)^{\phi} = \frac{w_t}{C_t^L} \tag{18}$$

$$K_{t+1}^L = (1-\delta) K_t^L + I_t^L \tag{19}$$

6.1.2 High-Income Households

The setup for high-income households is basically the same as for middle- and low-income households, except that high-income households do not receive government transfers Tr, as shown in the budget constraint (21).

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\ln C_t^H - \frac{(N_t^H)^{1+\phi}}{1+\phi} \right) \tag{20}$$

$$C_t^H + I_t^H + \delta K_t^H = (1-\tau_H) w_t N_t^H + r_t K_t^H \tag{21}$$

$$\frac{1}{C_t^H} = \beta \mathbb{E}_t \left[\frac{1}{C_{t+1}^H} (r_{t+1} + 1 - \delta) \right] \tag{22}$$

$$(N_t^H)^{\phi} = \frac{w_t}{C_t^H} \tag{23}$$

$$K_{t+1}^H = (1-\delta) K_t^H + I_t^H \tag{24}$$

6.1.3 Firms

Firms follow a Cobb–Douglas production function (25); technology follows an AR(1) stochastic shock process (26); the factor-demand optimality conditions equate marginal products with marginal factor rewards, as in (27) and (28); and the firm's marginal cost is defined in (29).

$$Y_t = A_t (N_t^L + N_t^H)^{\alpha} \cdot (K_t^L + K_t^H)^{1-\alpha} \tag{25}$$

$$\log A_t = \rho_A \log A_{t-1} + \varepsilon_t^A \tag{26}$$

$$w_t = \alpha \cdot \frac{Y_t}{N_t^L + N_t^H} \tag{27}$$

$$r_t = (1-\alpha) \cdot \frac{Y_t}{K_t^L + K_t^H} \tag{28}$$

$$MC_t = \frac{w_t (N_t^L + N_t^H) + r_t (K_t^L + K_t^H)}{Y_t} \tag{29}$$

6.1.4 Government

We assume that government revenue comes only from taxes on the two household types, and that government spending consists only of government purchases and transfers; the budget constraint is (30). Transfers follow an AR(1) stochastic shock process (31). Monetary policy is not considered for now.

$$G_t + Tr_t = \tau_L (w_t N_t^L + r_t K_t^L) + \tau_H (w_t N_t^H + r_t K_t^H) \tag{30}$$

$$\log Tr_t = \rho_{Tr} \log Tr_{t-1} + \varepsilon_t^{Tr} \tag{31}$$

6.1.5 The New Keynesian Phillips Curve

The New Keynesian Phillips curve (NKPC) describes the relationship between inflation and the firm's marginal cost (32); the full Calvo price stickiness mechanism is not considered, and a simplified price-adjustment ratio is used instead (33).

$$\pi_t = \beta \mathbb{E}_t [\pi_{t+1}] + \kappa (MC_t - 1) \tag{32}$$

$$\kappa = \frac{(1-\theta)(1-\beta\theta)}{\theta} \tag{33}$$

6.1.6 Resource Constraint

Market clearing requires:

$$Y_t = C_t^L + C_t^H + I_t^L + I_t^H + G_t \tag{34}$$

6.1.7 Impulse Response Analysis

The parameters of the model are set as in Table 4, with standard parameters following mainstream conventions in the literature (Li et al., 2024; Wang and He, 2017).

Table 4: Parameter calibration

β	ϕ	δ	α	τ_L	τ_H	θ	ρ_{Tr}	ρ_A	κ
0.96	1	0.025	0.33	0.1	0.25	0.75	0.9	0.9	0.093

The model is simulated using Dynare (Adjemian et al., 2024). Figure 4 shows the impulse responses of the model variables to a random shock to transfers; the horizontal axis of each subplot is the horizon after the shock, and the vertical axis is the standardized deviation from steady state. Faced with a positive transfer shock — that is, the government increasing redistribution toward middle- and low-income households — the consumption structure and total output adjust markedly, in turn relieving deflationary pressure. First, the shock raises the disposable income of middle- and low-income households directly by raising the wage w, stimulating their consumption C_L to rise rapidly; at the same time, because high-income households do not benefit directly from the shock, their consumption C_H does not expand to the same degree, although it does rise somewhat with the higher aggregate price level.

On the labor supply side, because the expansion of transfers depends on the revenue raised by higher taxes, households reduce labor supply N_L and N_H very slightly and rely more on capital income. Because tax rates differ, high-income households reduce labor supply more, while middle- and low-income households face stronger labor supply stickiness. On the capital side, because the total income of middle- and low-income households improves, they have a stronger incentive to supply capital K_L, which converts into investment I_L.

Turning to the variables most closely linked to deflation, the transfer shock expands total output Y, raises the wage w, and lowers the interest rate r, thus raising the price level through both expanded consumption and stimulated investment, and relieving deflationary pressure.

The model thus supports the view that improving the income distribution structure by raising the income share of middle- and low-income groups can, through both demand-driven and cost-push channels, push the price level upward and provide a hedging path for a deflationary environment.

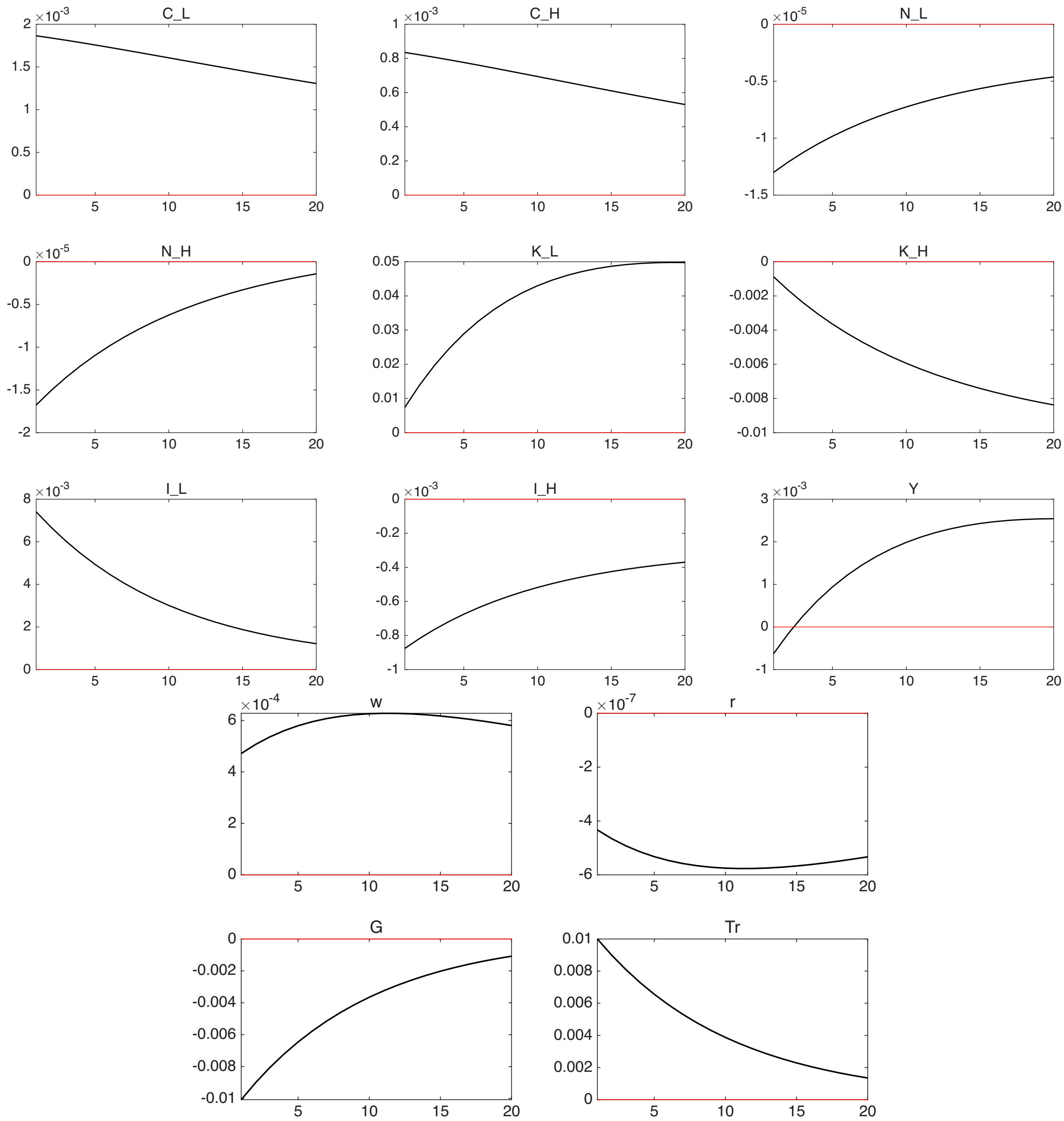


Figure 4: Impulse responses to a transfer-payment shock.

7 Conclusion and Outlook

7.1 Main Findings

This essay first assesses the current economic situation. Based on indicators such as the CPI and the PPI, it argues that since the third quarter of 2022 the Chinese economy has been facing a severe deflationary challenge, and then briefly catalogs the multiple negative effects of malignant deflation on the economy and its long-run damage to the growth mechanism. Combining the discussion of the causes of malignant deflation with the discussion of China's development path, the essay argues that a weak consuming class, high export dependence, an oversized real estate sector, and entrenched implicit debt are the direct causes of the current deflation, while the insufficient domestic demand caused by long-run distributional imbalance is the fundamental cause. To test this theory, the essay carries out simplified mathematical derivations, argues that improving the distribution structure helps relieve deflationary pressure, and provides empirical evidence using data on 77 economies from 2011 to 2021. The essay also builds a simplified three-sector New Keynesian DSGE model with heterogeneous households for policy simulation, and finds that raising the income share of middle- and low-income groups can push the price level back up through both demand-driven and cost-push channels.

7.2 Social Security, Labor Law, and the Future of Our Economy

This essay has no intention of judging history, nor is it able to offer implementable solutions. Finding causes is always easy; proposing remedies is the real difficulty. In the analysis above we pointed out that the Chinese economy's long-standing dependence on investment and exports rests on the sustained suppression of labor costs, and that workers have long occupied a structurally disadvantaged position of lagging income growth and weak social security. This distorted distributional pattern not only weakens the consumption capacity of the household sector; it also makes the whole society more prone to falling into a deflationary spiral when growth slows. To break this structural impasse, the key is to rebuild workers' rights protection and income confidence, which must start from strict enforcement of labor law, improved labor compensation, and a sounder social security system.

First, the eight-hour working day and the related working-hour protection rules should be strictly implemented. Compressing overtime and cracking down on "forced overtime" frees up the jobs that overtime had absorbed, while improving workers' physical and mental health. A reasonable allocation of working time not only helps expand employment; it also frees up time for leisure consumption, activating service-sector demand such as tourism, culture, education, and fitness, and providing long-term support for the consumption structure.

Second, the laws and regulations on organizing trade unions and safeguarding their substantive functions should be truly enforced. For a long time, insufficient union coverage and hollow consultation mechanisms have severely constrained workers' bargaining power, leaving wage growth persistently lagging productivity growth. The construction of grassroots unions should be vigorously promoted, collective wage consultation should be implemented, and a stable, predictable labor-management consultation mechanism should be built to reshape the income distribution structure at the institutional level.

This process will inevitably raise labor costs. To prevent the employment pressure on firms from being passed on to prices or jobs, the government should establish supporting adjustment mechanisms: for example, setting up a dedicated redistribution fund to support small and medium-sized enterprises in raising their wage-paying capacity through structural transfers and special government bonds; simultaneously advancing the reform of corporate profit redistribution, increasing the share of profits distributed to workers, and strengthening the link between wages and firm performance; and appropriately raising the marginal tax rate on high-income groups to balance the tax structure and improve the overall efficiency of redistribution.

At the same time, the imbalance in social security capacity between urban and rural areas is an important cause of chronic consumption weakness and distorted population mobility. The service capacity of rural pension insurance and medical insurance should be upgraded — not only to raise the level of protection itself, but also to strengthen the security expectations and confidence in life of the rural resident population. It is recommended that a unified urban–rural basic security floor be established, cross-regional settlement of medical insurance and national pooling of pensions be promoted, and the urban–rural gap in basic welfare be genuinely narrowed.

Finally, a "de-real-estate-binding" model of social governance should be promoted to rebuild the livability of rural areas. By increasing fiscal investment in rural infrastructure, public services, and township enterprises, key public services such as education, medical care, employment, and social security would no longer be tightly tied to urban homeownership, allowing people to enjoy basic living conditions without buying a house. This policy turn that deconstructs "urban-centrism" would fundamentally repair the structural dislocation of the current "population—housing—debt" chain, creating a more friendly institutional environment for labor mobility and fertility intentions.

Structural reform is usually thought to work slowly. In fact, merely sending the market clear and reliable signals of labor protection and rural reconstruction is enough to generate positive policy spillovers in the short run through the expectation mechanism.

In short, only an institutional system based on labor protection, income security, and social fairness can truly sustain a consumption-driven domestic demand structure and guide the economy out of the structural trap of deflation. Labor is a factor of production, but workers are the true agents of the economy and society. If a country cannot guarantee its workers decent jobs and fair distribution, it will hardly be able to build a stable consumption base and healthy market expectations. The reconstruction of the labor system is not only a correction of historical imbalance, but also a restart of future economic vitality.

This essay contains no original contribution; it is merely an exercise in which the author sorts out personal thoughts and practices economic techniques, and at most it can serve as a memorial to these dull times.[4]

> Лишь мы, работники всемирной
> Великой армии труда
> Владеть землёй имеем право,
> Но паразиты — никогда!

[4]The two epigraphs that close this essay are, respectively, an excerpt from *The Internationale* and an excerpt from the song "Prekrasnoe dalyoko" (Beautiful Far Away). Translation of the first: "Only we, the workers of the world's great army of labor, have the right to own the land — but the parasites, never!" Translation of the second: "Beautiful far away, do not be cruel to me, do not be cruel to me, do not be cruel. From a pure source toward the beautiful far away, toward the beautiful far away I begin my journey."

Прекрасное далёко, не будь ко мне жестоко,
Не будь ко мне жестоко, жестоко не будь.
От чистого истока в прекрасное далёко,
В прекрасное далёко я начинаю путь.

— Интернационал — Прекрасное далёко

References

Adjemian, S., Juillard, M., Karamé, F., Mutschler, W., Pfeifer, J., Ratto, M., Rion, N., Villemot, S., 2024. Dynare: Reference Manual, Version 6. Dynare Working Papers 80. CEPREMAP.

Chen, X., Liu, L., Xiao, Z., Chen, Y., 2021. Differences in the determinants of producer-sector deflation and economy-wide deflation: A new perspective from machine learning (in chinese). China Industrial Economics , 26–44.

Commission on Growth, W.B., 2008. The growth report: Strategies for sustained growth and inclusive development. World Bank Publications.

Guiso, L., Sapienza, P., Zingales, L., 2015. The value of corporate culture. Journal of Financial Economics 117, 60–76.

Hunter, J.D., 2007. Matplotlib: A 2d graphics environment. Computing in Science & Engineering 9, 90–95.

Li, J., Wang, T., Su, Z., 2024. Optimal monetary policy under digital technology shock. Technological Forecasting and Social Change 200, 123133.

Tang, Y., Zhang, G., 2020. Excess money issuance and deflation: A theoretical review and factual evidence on china's monetary puzzle (in chinese). Nanjing Journal of Social Sciences , 29–36, 46.

Wang, X., He, J., 2017. The impact of technological progress on deflationary expectations (in chinese). Research on Economics and Management 38, 3–13.

Xia, G., 2023. The intellectual origins, basic concepts, and historical experience of deflation theory: A theoretical review (in chinese). Cai Jing Zhi Ku (Finance Think Tank) 8, 67–86, 145.

Zhu, T., 2019. How does real-economy debt affect the stability of the financial system? theoretical mechanisms and an interpretive framework (in chinese). Chinese Review of Financial Studies 11, 25–37, 123.